# Band-1 Hadamard Order 9884 – Classical Paley-I (q=9883); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9884 Paley-I PASS; Paley-II FAIL/SKIP (q=4941=3^4*61 NOT prime / not prime power; ===1 mod 4); Kronecker FAIL/SKIP (4*2471; 2471===3 mod 4 NOT admissible; NOT in ONBOX_SEEDS); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

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2026-09-23

**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
**Parents.** `2026-09-23_Band1_Hadamard_9872_PaleyI`; `2026-09-23_Band1_Hadamard_9860_PaleyI`; `2026-09-22_Band1_Hadamard_9852_PaleyI`.
**Lab.** `/workspace/experiments/band1/hadamard_9884_paleyI/`.
**Cutoff.** 2026-09-23T09:05:12Z (~5:05AM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite construction of order 9884 only (verified max|HH^T-nI|=0), via classical Paley-I q=9883 (prime ===3 mod 4). 9884 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9884: Paley-I **PASS** ($q = 9883$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 4941 = 3^4 \cdot 61$ NOT prime / not prime power; $\equiv 1 \pmod 4$); Kronecker **FAIL/SKIP** (note $9884 = 4 \cdot 2471$; $2471 \equiv 3 \pmod 4$ NOT admissible; NOT in ONBOX_SEEDS). **9884** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9872+9860+9852+9840+9812+9804+9792+9788+9768+9744+9740+9720+9680+9644+9632+9624-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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## Abstract

Finite construction of order **9884**.

- $H_{9884}$ = Paley-I ($q = 9883$) – classical SPK

- Paley-II for 9884: **FAIL/SKIP** ($q = 4941 = 3^4 \cdot 61$ NOT prime / not prime power; $\equiv 1 \pmod 4$)
- Kronecker for 9884: **FAIL/SKIP** ($4 \cdot 2471$; $2471 \equiv 3 \pmod 4$ NOT admissible; NOT in ONBOX_SEEDS)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9884.py
```

**Results.** overall_pass=True; units 172/172; row blocks 78/78 (block size 128); wall 787.2895s; constructed max|HHˆT-nI|=0.

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## Arithmetic

- Target $n = 9884 = q + 1$ with $q = 9883$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 4941 = 3^4 \cdot 61$ NOT prime / not prime power; $\equiv 1 \pmod 4$ — FAIL/SKIP.
- Kronecker: $9884 = 4 \cdot 2471$; $2471 \equiv 3 \pmod 4$ NOT admissible; NOT in ONBOX_SEEDS — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_9884 Paley-I | Paley-I q=9883 | 0 | 769.8197 |
| H_9884 Paley-II | FAIL/SKIP (q=4941=3ˆ4*61 NOT prime / not prime power; ===1 mod 4) | – | – |
| H_9884 Kronecker | FAIL/SKIP (4*2471; 2471===3 mod 4 NOT admissible; NOT in ON-BOX_SEEDS) | – | – |

## SPK closure note

Seeds include prior SPK through **9872** lineage (**9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; ...; **NOT** 8948/8644/8516/...) + verified **9884** (+ prior **9872+9860+9852+9840+9812+9804+9792+9788+9768+9744**). Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 9884 in SPK | **True** (Paley-I q=9883) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q=9887->n=9888 Paley-I DIRECT q=9887 OR q=9907->n=9908 Paley-I DIRECT q=9907 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=9923->n=9924 Paley-I DIRECT q=9923 OR Track B / Object 1 meter await